

STAT 520

Applied Multivariate Analysis

SIMULTANEOUS INFERENCE

Bonferroni confidence sets and confidence ellipsoid

```
Data SWEAT;          /* # 5.7, p.261. Sweat data from Example 5.2. */
infile "C:\Users\baron\520\SAS and data\Johnson Wichern Data sets\T5-1.dat";
input Rate Sodium Potassium; run;

PROC MEANS DATA=SWEAT N MEAN CLM ALPHA=0.0167; VAR Sodium;
/* CLM asks to compute (1-0.0167) confidence interval, where 0.0167=0.05/3 for p=3 */
OUTPUT OUT=Stats1 LCLM=LCL_Sodium UCLM=UCL_Sodium; run; /* Save confidence limits */
```

	N	Mean	Lower 98.33% CL for Mean	Upper 98.33% CL for Mean
	20	45.4000000	37.1060295	53.6939705

```
PROC MEANS DATA=SWEAT N MEAN CLM ALPHA=0.0167; VAR Potassium;
OUTPUT OUT=Stats2 LCLM=LCL_Potassium UCLM=UCL_Potassium; run;

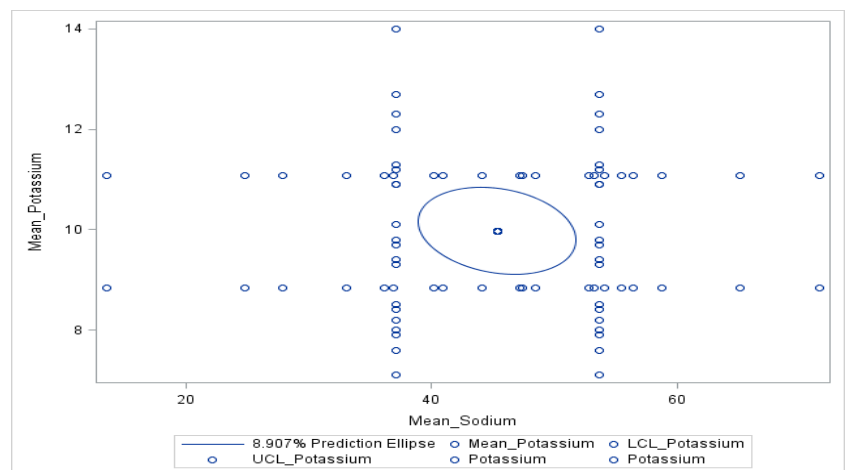
DATA Stats; MERGE Stats1 Stats2; DO I=1 TO 20; OUTPUT; END; run; /* Duplicate rows */
```

```
DATA SWEAT; MERGE SWEAT Stats;
N=20; P=3; alpha=0.05;
CriticalF = (((N-1)*P)/(N-P))*FINV(1-alpha, P, N-P)/N;
LevelForChisq = 1-PROBCHI(CriticalF,P); run;
```

```
PROC PRINT DATA=SWEAT; run;
```

LCL_Sodium	UCL_Sodium	LCL_Potassium	UCL_Potassium	CriticalF	LevelForChisq
37.1060	53.6940	8.84739	11.0826	0.53593	0.91093

```
PROC SGPLOT DATA=SWEAT;
scatter X=Sodium Y=Potassium;
scatter X=Sodium Y=LCL_Potassium;
scatter X=Sodium Y=UCL_Potassium;
scatter X=LCL_Sodium Y=Potassium;
scatter X=UCL_Sodium Y=Potassium;
ellipse X=Sodium Y=Potassium /
alpha=0.91093;
run;
```



Simultaneous hypothesis testing. Adjustments for multiplicity.

```
Data SWEAT; set SWEAT;
Y1=RATE-4; Y2=SODIUM-50; Y3=POTASSIUM-10; run; /* Now we just test Ho: MU = 0 */

PROC MEANS DATA=SWEAT PROBT; VAR Y1; OUTPUT OUT=SWEAT_PVAL1 PROBT=raw_p; run;
/* Save p-values from standard two-sided t-tests */
```

```
PROC MEANS DATA=SWEAT PROBT; VAR Y2; OUTPUT OUT=SWEAT_PVAL2 PROBT=raw_p; run;
PROC MEANS DATA=SWEAT PROBT; VAR Y3; OUTPUT OUT=SWEAT_PVAL3 PROBT=raw_p; run;
DATA SWEAT_PVAL; SET SWEAT_PVAL1 SWEAT_PVAL2 SWEAT_PVAL3; run;

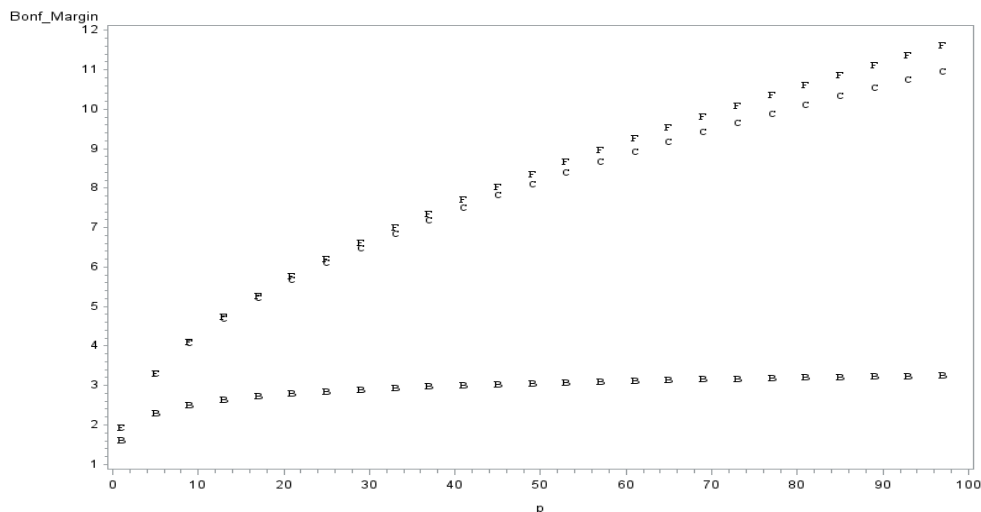
PROC MULTTEST PDATA=SWEAT_PVAL BON HOLM HOMMEL SIDAK; RUN;
```

Test	p-Values				
	Raw	Bonferroni	Stepdown Bonferroni	Sidak	Hommel
1	0.1080	0.3240	0.3240	0.2903	0.2428
2	0.1619	0.4856	0.3240	0.4113	0.3238
3	0.9354	1.0000	0.9354	0.9997	0.9354

Compare margins of simultaneous confidence intervals

```
DATA MARGINS; alpha=0.05; N=1000;
DO p=1 to 100 by 4;
  Bonf_Margin = TINV(1-alpha/p,n-1);
  T2_Margin = sqrt( p*(n-1)/(n-p)*FINV(1-alpha,p,n-p) );
  Chisq_Margin= sqrt( CINV(1-alpha,p) );
  OUTPUT; /* This is needed to save results for each p */
END;

PROC GGPLOT DATA=MARGINS;
PLOT Bonf_Margin*p='B' T2_Margin*p='F' Chisq_Margin*p='C' / OVERLAY; run;
```



Fun fact: more sodium, less potassium when sweating faster

```
PROC SGPLOT DATA=SWEAT;
scatter X=Rate Y=Sodium;
scatter X=Rate Y=Potassium;
ellipse X=Rate Y=Sodium;
ellipse X=Rate Y=Potassium;
run;
```

